



Q1 Keyword Arguments

5 marks

Write a function employee(empid, name, salary) and call the function by changing the order of the arguments using keywords.

Your Code

```
1 def function(empid,name,salary):  
2     print("Emp id ",empid)  
3     print("name",name)  
4     print("salary",salary)  
5 function(empid = 2736,name = "Revanthsai",salary = 5473635)
```

Input

Output

```
Emp id 2736  
name Revanthsai  
salary 5473635
```



Q2 Keyword Arguments

5 marks

Write a function book(title, author, price) and display the book details using keyword arguments.

Your Code

```
1 def function(title,author,price):  
2     print("title of the book is ",title)  
3     print("author of book",author)  
4     print("price of book",price)  
5 function(title = "The tale of naruto Uzumaki ",author = "Jiraya", price = 54321)
```

Input

stdin input

Output

```
title of the book is The tale of  
naruto Uzumaki  
author of book Jiraya  
price of book 54321
```



Q3 Variable-Length Arguments (*args)

5 marks

Write a function that accepts any number of integers and returns their sum.

Your Code

```
1 def sum(*numbers):  
2     total = 0  
3     for i in numbers:  
4         total = total + i  
5     return total  
6 print("sum=", sum(10,20,20,30,40,10))  
7  
8
```

Input

Output

```
sum= 130
```



Q4 Variable-Length Arguments (*args)

5 marks

Write a function that accepts any number of marks and returns the average.

Your Code

```
1 def avg(*marks):  
2     total = 0  
3     count = 0  
4     for i in marks:  
5         total = total + i  
6         count = count + 1  
7         average = total / count  
8         return average  
9 print("average", avg(3, 3, 2, 1, 4, 5))  
10  
11  
12
```

Input

stdin input

Output

average 3.0



Q5 Variable-Length Arguments (*args)

5 m

Write a function that accepts any number of strings and prints each string on a new line.

Your Code

```
1 def print_strings(*args):  
2     for s in args:  
3         print(s)  
4  
5 # Example  
6 print_strings("Apple", "Banana", "Cherry", "Orange")
```

Input

stdin input

Output

```
Apple  
Banana  
Cherry  
Orange
```



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Q6 Variable-Length Arguments (*args)

5 marks

Write a function that accepts any number of numbers and returns the largest value.

Your Code

```
1 def largest(*numbers):  
2     return max(numbers)  
3  
4 # Example  
5 print(largest(10, 25, 7, 40, 18))
```

Input

stdin input

Output

40

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Q7 Recursive Functions

5 marks

Write a recursive function to generate the Fibonacci series up to n terms.

Your Code

```
1 def fibonacci(n):
2     if n <= 1:
3         return n
4     return fibonacci(n - 1) + fibonacci(n - 2)
5
6 n = int(input("Enter number of terms: "))
7
8 print("Fibonacci Series:")
9 for i in range(n):
10     print(fibonacci(i), end=" ")
```

Input

7

Output

```
Enter number of terms: Fibonacci
Series:
0 1 1 2 3 5 8
```



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Q8 Recursive Functions

5 marks

Write a recursive function to find the sum of digits of a number.

Your Code

```
1 def sum_digits(n):  
2     if n == 0:  
3         return 0  
4     return (n % 10) + sum_digits(n // 10)  
5  
6 num = int(input("Enter a number: "))  
7 print("Sum of digits =", sum_digits(num))
```

Input

123

Output

Enter a number: Sum of digits = 6



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Q9 Recursive Functions

5 marks

Write a recursive function to reverse a string.

Your Code

```
1 def reverse_string(s):  
2     if len(s) == 0:  
3         return s  
4     return reverse_string(s[1:]) + s[0]  
5  
6 text = input("Enter a string: ")  
7 print("Reversed string:", reverse_string(text))
```

Input

Hello

Output

```
Enter a string: Reversed string:  
olleH
```